

The Exact Nonproperness Locus of the Quadratic-Gauge Keller Maps

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Abstract

For the all-degree quadratic-gauge Keller maps, we identify the reduced nonproperness locus with the discriminant of the scalar inverse equation. We compute the complete fiber table on the degenerate slice $\Pi = 0$ and the exact global Π -adic order of the discriminant. These geometric results are independent of the prescribed finite-étale-fiber theorem.

1 The quadratic-gauge input

Let K be a characteristic-zero field and

$$G(S) = g_1 S + g_2 S^2 + \cdots + g_N S^N, \quad g_1 g_3 g_N \neq 0. \quad (1.1)$$

The quadratic-gauge map $F_G = (\Pi, B, C) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is the map constructed in the companion prescribed-fiber paper. Its scalar inverse equation is

$$E_{\Pi, B, C}(S) = G_{\Pi}(S) - \frac{g_1}{2}(BS^2 + C),$$

where

$$G_{\Pi}(S) = g_1 S + \Pi(g_2 S^2 + g_3 S^3) + \sum_{k=4}^N g_k \Pi^k S^k.$$

It satisfies

$$\det DF_G = -2, \quad \text{gdeg}(F_G) = N.$$

For $\pi \neq 0$, its literal fiber is

$$F_G^{-1}(\pi, b, c) \simeq \text{Spec} \left((K[S]/(E_{\pi, b, c}))[1/\overline{E'_{\pi, b, c}}] \right). \quad (1.2)$$

These identities follow from the compact inverse design and root-to-source reconstruction in the prescribed-fiber paper; the present note begins with them and studies the boundary.

For a squarefree $P \in K[T]$ of degree $N \geq 3$ and $a \in K$ with $P'(a)P'''(a) \neq 0$, the prescribed-fiber seed is

$$G(S) = P(a + S) - P(a).$$

After the determinant-one output normalization, the distinguished target is

$$y_{P, a} = \left(1, 0, -\frac{2P(a)}{P'(a)} \right), \quad (1.3)$$

and its inverse equation is $P(a + S)$.

2 The exact nonproperness locus

The prescribed-fiber realization theorem does not require the calculation below. This companion note gives an intrinsic geometric description of fullness for the quadratic gauge.

Work geometrically over $\overline{K}$. For a polynomial map $F : \mathbb{A}_{\overline{K}}^n \rightarrow \mathbb{A}_{\overline{K}}^m$, let S_F be the set of target points at which F is not finite over any Zariski neighborhood. Since source and target are affine, this is the same as failure of properness over every neighborhood. Equivalently, compactify the graph in $\mathbb{P}_{\overline{K}}^n \times \mathbb{A}_{\overline{K}}^m$; then S_F is the image of its boundary. We give S_F its reduced closed-subscheme structure. Over $\mathbb{C}$ this agrees with the usual analytic Jelonek set, defined by sequences escaping to infinity.

For a polynomial h , write $V_{\text{red}}(h)$ for the reduced closed subscheme $V(\sqrt{(h)})$. Put

$$\Delta_G(\Pi, B, C) = \text{Disc}_S(E_{\Pi, B, C}(S)). \quad (2.1)$$

Here the discriminant is the universal degree- N polynomial expression in the coefficients of $E_{\Pi, B, C}$. We do not replace it on a special slice by the discriminant of the lower-degree specialization.

Lemma 2.1 (Base change of the graph-boundary locus). *Let $k \subset k'$ be a field extension and let $F : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be a polynomial map. Let $\overline{\Gamma}_F$ be the scheme-theoretic closure of its graph in $\mathbb{P}_k^n \times \mathbb{A}_k^m$, put*

$$Z_F = \overline{\Gamma}_F \cap ((\mathbb{P}_k^n \setminus \mathbb{A}_k^n) \times \mathbb{A}_k^m),$$

and give the image of the proper morphism $Z_F \rightarrow \mathbb{A}_k^m$ its reduced closed-subscheme structure S_F . Then

$$Z_{F_{k'}} = Z_F \times_k k', \quad |S_{F_{k'}}| = |(S_F)_{k'}|.$$

In particular, whether a geometric target point lies in the graph-boundary locus is invariant under extension of algebraically closed fields.

Proof. Field extension is faithfully flat. On each standard affine chart of $\mathbb{P}_k^n \times \mathbb{A}_k^m$, the ideal of the graph closure is the saturation $I : x_0^\infty$ of the graph ideal by the equation x_0 of the hyperplane at infinity. Since the coordinate rings are noetherian, this saturation equals $I : x_0^r$ for some r ; flatness gives

$$(I : x_0^r) \otimes_k k' = (I \otimes_k k') : x_0^r.$$

Thus scheme-theoretic graph closure commutes with the field extension. Intersecting with $x_0 = 0$ also commutes with base change, which proves the identity for Z_F .

For the image statement, let y' be a point of $\mathbb{A}_{k'}^m$, with image y in $\mathbb{A}_k^m$. The fiber $(Z_F)_{y'}$ is obtained from $(Z_F)_y$ by a field extension. A finite-type scheme over a field is empty if and only if it becomes empty after a field extension. Hence $(Z_F)_{y'}$ is nonempty exactly when $(Z_F)_y$ is nonempty. Since a proper morphism has closed image, these fiberwise equivalences give the asserted equality of underlying closed sets. $\square$

Lemma 2.2 (Fiber cardinality and nonproperness in characteristic zero). *Let k be an algebraically closed field of characteristic zero and let $F : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be a dominant étale polynomial map of geometric degree d . Then*

$$y \notin S_F \iff \#F^{-1}(y) = d. \quad (2.2)$$

Proof. Jelonek proves this multiplicity criterion for complex polynomial maps [1, Proposition 6]. We record why that stated complex result is sufficient here. Since F is étale, its Jacobian determinant is a nonzero unit of $k[x_1, \dots, x_n]$, hence a nonzero constant. Descend the coefficients of F , this constant Jacobian identity, and the coordinates of y to a finitely generated field $k_0/\mathbb{Q}$. The resulting

map F_0 is étale and therefore dominant; its geometric degree is still d . We also obtain $y_0 \in \mathbb{A}^n(k_0)$ whose base changes are y .

Choose an embedding $k_0 \hookrightarrow \mathbb{C}$ and an algebraic closure Ω of k_0 . The two embeddings of k_0 , into k and into $\mathbb{C}$, extend to embeddings

$$\Omega \hookrightarrow k, \quad \Omega \hookrightarrow \mathbb{C}.$$

Lemma 2.1 shows that membership of y_0 in the graph-boundary locus has the same truth value after each of these scalar extensions. The fiber of an étale morphism is finite and reduced; its number of geometric points is unchanged by an extension of algebraically closed fields. The degree of the generic finite separable extension is likewise invariant under scalar extension. We may therefore test both sides of (2.2) after base change to $\mathbb{C}$. There Proposition 6 of [1], with every local multiplicity equal to one, gives the equivalence. $\square$

Lemma 2.3 (Exact Π -adic discriminant order). *Assume $N \geq 4$, put $r = N - 3$, and let*

$$R = \overline{K}[B, C], \quad L = \text{Frac}(R).$$

For the degree- N polynomial $E = E_{\Pi, B, C}(S)$, viewed in $L((\Pi))[S]$, extend $\text{ord}_{\Pi}(\Pi) = 1$ to an algebraic closure of $L((\Pi))$. The roots split into three Newton blocks:

number of roots	$\text{ord}_{\Pi}(s)$	residual polynomial	
2	0	$BZ^2 - 2Z + C$	
1	-1	$Z - \frac{g_1 B}{2g_3}$	(2.3)
r	$-\frac{r+2}{r}$	$g_3 + g_N Z^r$.	

All three residual polynomials are separable over L . Consequently

$$\text{ord}_{\Pi} \text{Disc}_S(E) = N^2 - 3N - 2,$$

and, in $R[\Pi]$,

$$\text{Disc}_S(E) = \Pi^{N^2 - 3N - 2} D_G, \quad D_G(0, B, C) = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} B^2 (1 - BC).$$

Proof. The coefficient valuations give the lower Newton polygon

$$(0, 0) \longrightarrow (2, 0) \longrightarrow (3, 1) \longrightarrow (N, N).$$

Its slopes are 0, 1, and $(r+2)/r$, with horizontal lengths 2, 1, and r . Rescaling S along each edge gives exactly the three residual polynomials in (2.3). The first has discriminant $4(1 - BC) \neq 0$ in L , the second is linear, and the third is separable because $\text{char } L = 0$ and $g_3 g_N \neq 0$.

Write the roots in the three blocks as A_1, A_2, T , and $V_1, \dots, V_r$. Roots in different blocks have different valuations, so the valuation of their difference is the smaller of the two valuations. Within the first and third blocks, distinct roots of the residual polynomial have distinct initial terms. Thus

$$\text{ord}_{\Pi}(A_1 - A_2) = 0, \quad \text{ord}_{\Pi}(V_i - V_j) = -\frac{r+2}{r} \quad (i \neq j).$$

This is the precise use of residual separability. It proves equality in each entry of the following ledger:

pair type	number of pairs	$\text{ord}_\Pi(s_i - s_j)$	contribution to $\text{ord}_\Pi \prod_{i < j} (s_i - s_j)^2$
$A-A$	1	0	0
$A-T$	2	-1	-4
$A-V$	$2r$	$-(r+2)/r$	$-4(r+2)$
$T-V$	r	$-(r+2)/r$	$-2(r+2)$
$V-V$	$r(r-1)/2$	$-(r+2)/r$	$-(r-1)(r+2)$.

The leading coefficient of E is $g_N \Pi^N$, so its contribution to

$$\text{Disc}(E) = \text{lc}(E)^{2N-2} \prod_{i < j} (s_i - s_j)^2$$

is $2N(N-1)$. Adding the last column gives

$$2N(N-1) - 4 - 4(r+2) - 2(r+2) - (r-1)(r+2) = N^2 - 3N - 2.$$

It remains only to identify the first nonzero coefficient. Let α_1, α_2 be the roots of $BZ^2 - 2Z + C$, put $b_0 = g_1 B / (2g_3)$, and let $v_1, \dots, v_r$ be the roots of $g_3 + g_N Z^r$. Taking the initial term in every factor of the root-product formula gives

$$\begin{aligned} & \pm g_N^{2N-2} (\alpha_1 - \alpha_2)^2 b_0^4 \left(\prod_j v_j \right)^6 \prod_{i < j} (v_i - v_j)^2 \\ & = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} B^2 (1 - BC), \end{aligned}$$

because

$$(\alpha_1 - \alpha_2)^2 = \frac{4(1 - BC)}{B^2}, \quad \left(\prod_j v_j \right)^6 = \left(\frac{g_3}{g_N} \right)^6, \quad \prod_{i < j} (v_i - v_j)^2 = \pm r^r \left(\frac{g_3}{g_N} \right)^{r-1}.$$

This coefficient is nonzero in R . Since Π -adic order in the domain $R[\Pi]$ is unchanged after passage to $L[\Pi]$, the asserted factorization and specialized coefficient follow. $\square$

Theorem 2.4 (Exact reduced quadratic-gauge nonproperness). *For every seed (1.1),*

$$\boxed{S_{F_G} = V_{\text{red}}(\Delta_G)}. \quad (2.4)$$

Equivalently, their sets of geometric points satisfy $|S_{F_G}| = V_{\overline{K}}(\Delta_G)$. More precisely, the following statements hold.

1. *If $\pi \neq 0$ and the multiplicity partition of $E_{\pi, b, c}$ is $(m_1, \dots, m_s)$, then*

$$\#F_G^{-1}(\pi, b, c) = \#\{i : m_i = 1\}, \quad N - \#F_G^{-1}(\pi, b, c) = \sum_{m_i \geq 2} m_i. \quad (2.5)$$

2. *On $\Pi = 0$, the complete geometric fiber table is*

target stratum	$\#F_G^{-1}(0, B, C)$	lost sheets
$B \neq 0, BC \neq 1$	3	$N - 3$
$B \neq 0, BC = 1$	1	$N - 1$
$B = 0$	1	$N - 1$.

(2.6)

3. For $N = 3$, the raw resultant has an extraneous leading-coefficient factor:

$$\text{Res}_S(E, E') = \text{unit} \cdot \Pi \Delta_G, \quad \Delta_G(0, B, C) = \frac{g_1^4}{4} B^2(1 - BC).$$

Hence the entire plane $\Pi = 0$ is not a nonproperness component in degree three.

4. For $N \geq 4$, put

$$e_N = N^2 - 3N - 2.$$

Then

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2(1 - BC), \quad \kappa_G \in K^*. \quad (2.7)$$

In particular,

$$S_{F_G} = V_{\text{red}}(\Pi D_G),$$

whose underlying set is $V(\Pi) \cup V(D_G)$, and the resultant and discriminant define the same reduced closed subscheme.

Proof. Because F_G is étale, every geometric fiber is finite and reduced. Its generic cardinality is N , so Lemma 2.2 applies with $d = N$.

Suppose first that $\pi \neq 0$. The localized reconstruction (1.2) identifies the fiber with

$$\text{Spec}((\overline{K}[S]/(E_{\pi,b,c}))[1/\overline{E'_{\pi,b,c}}]).$$

Thus a distinct root contributes one affine point exactly when it is simple. This proves (2.5); it also shows that the fiber has cardinality N exactly when $\Delta_G(\pi, b, c) \neq 0$.

It remains to analyze $\Pi = 0$, where the inverse chart used above is not available. In the source, $\Pi = tq$. On $q = 0$, one has

$$B = y, \quad C = 2S - BS^2, \quad S = \frac{x}{t}.$$

The residual inverse equation is therefore

$$BS^2 - 2S + C = 0. \quad (2.8)$$

A root reconstructs on $q = 0$ precisely when $1 - BS \neq 0$. Hence this divisor contributes two points when $B \neq 0$ and $BC \neq 1$, no point when $B \neq 0$ and $BC = 1$, and one point when $B = 0$.

On $t = 0$, the relation $1 + xy = 0$ gives $y = -1/x$, while

$$q = \frac{g_1}{g_3} y^2, \quad B = -2y = \frac{2}{x}.$$

Thus $t = 0$ contributes one point when $B \neq 0$ and none when $B = 0$. Once x and y are fixed, the equation for C is linear in z with nonzero coefficient, so the point is unique. The divisors $t = 0$ and $q = 0$ do not meet. This proves (2.6).

For $N = 3$, specializing the degree-three discriminant at $\Pi = 0$ gives

$$\Delta_G(0, B, C) = \frac{g_1^4}{4} B^2(1 - BC).$$

Its zero set is exactly the two short-fiber strata in (2.6). Since the leading coefficient is $g_3 \Pi$, the resultant has one additional factor Π .

Now let $N \geq 4$ and put $r = N - 3$. Lemma 2.3 gives the global polynomial identity

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2(1 - BC), \quad \kappa_G = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} \neq 0.$$

In particular, e_N is the exact global Π -adic order. After specializing (B, C) , the pointwise order is e_N when $B(1 - BC) \neq 0$ and may jump on the two displayed special strata.

No general assertion about discriminants under a drop of leading coefficient is being used. For $N = 3$, the explicit specialized formula cuts out exactly the two short-fiber strata. For $N \geq 4$, Δ_G vanishes on the whole slice $\Pi = 0$, and every fiber on that slice has fewer than N points by (2.6). On $\Pi \neq 0$, the ordinary degree- N discriminant detects exactly the repeated-root, hence short-fiber, locus. The fiber table and (2.2) now show that the zero set of Δ_G is exactly the underlying nonproperness set, and taking radicals gives (2.4). $\square$

Corollary 2.5 (Intrinsic fullness of the distinguished fiber). *For the seed $G(S) = P(a + S) - P(a)$ associated with a squarefree polynomial presentation, the distinguished target (1.3) lies in the maximal open subset on which F_G is finite étale of degree N .*

Proof. At $y_{P,a}$, the inverse polynomial is $P(a + S)$. Hence

$$\Delta_G(y_{P,a}) = \text{Disc}(P(a + S)) = \text{Disc}(P) \neq 0.$$

Theorem 2.4 places the target outside the nonproperness hypersurface. Proper and étale is finite étale, and the degree is N . $\square$

Verification status

The graph-boundary base-change argument, Jelonek descent, complete $\Pi = 0$ fiber table, and Newton-polygon discriminant calculation are ordinary mathematical proofs and are not part of the Lean formalization of the prescribed-fiber theorem. The exact symbolic audit is

`scripts/verify_quadratic_gauge_nonproperness.py`.

References

- [1] Z. Jelonek, *The set of points at which a polynomial map is not proper*, Ann. Polon. Math. **58** (1993), 259–266, doi:10.4064/ap-58-3-259-266.